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PISA's PL4 Format Frequency domain

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Binary file byte's positions for frequency domain simulations (UNDER DEVELOPMENT)

	"columns"																
"Row"	8 bytes								8 bytes								Notes
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
0																	ASCII HEADER
1																	Seems to be some kind of versioning
2	00	00	00	00	?FMAX?				DELTA F								
3	#VAR X2								PL4 SIZE+1								
4	CPX				NCH				00	00	00	00	00	00	00	00	"Row" of zeroes before var names
5	20	20	20	VT	Node From				Node To								Begins variables, one for each "row", ascii encoding.
...																	
5+n	20	20	20	VT	Node From				Node To								
5+n+1	t[0]				var1[0]				var2[0]				var3[0]				Begins actual data, sequence of IEEE 754 float in little endian order
...																	

Notes:

- FMAX - This field is related to FMAX, however it is not exactly TMAX. *Still needs further investigation*;
- DELTA F - 4 bytes IEEE 754 floating point, little endian;
- #VAR X2 - In this field there is twice the number of variables. Not including the time. (4 bytes integer);
- PL4 SIZE+1 - The PL4 file size in bytes + 1 byte (4 bytes integer);
- In 'row' 4 we have CPX / NVAR both (4 bytes integer);

Description	CPX	NCH	ORDER
Nothing	0	1	-
Polar	1	2	Abs / Angle
Rectangular	2	2	Re / Im
Rectangular and Polar	3	4	Abs / Angle / Re / Im

- In 'row' 5 begins the variables description, where:
 - Every variable 'row' starts with the sequence of 202020 in the first three bytes;
 - The following byte 'VT', or fourth 'column', contains the variable type (ascii encoding) - (See [Variable Types](#));
 - The next 6 bytes are the "Node From" (ascii encoding);

- The next 6 bytes are the "Node To" (ascii encoding);
- In 'row' 5 + the number of variables, the actual data is recorded:
 - They are composed of 4 bytes fields for IEEE 754 floating point, little endian;
 - The first 4 bytes are the time for the first simulation step;
 - Then comes the variables data, in the same sequence which the variables are given, for the first time step;
 - The above mentioned sequence repeats for all simulation's steps.
- With the current information, the number of simulations steps can be calculated as: $STEPS = (PL4\ SIZE - 5 * 16 - \#VAR * 16) / ((\#VAR + 1) * 4)$;
 - This information allows to dynamically allocate the array to store the simulations results with the dimensions of $STEPS \times \#VAR + 1$ (the +1 stands for the time column);

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